

TIME SERIES ANALYSIS & CLASSIFICATION

2025/2026

1 Instructors Information

Tarek Medkour

Professor of Statistics

E-mail: tarek.medkour@ensia.edu.dz

Houssam Brairi

Assistant Professor of Statistics

E-mail: houssam.brairi@ensia.edu.dz

Oualid Ouarem

Assistant Professor of Computer Science

E-mail: oualid.ouarem@ensia.edu.dz

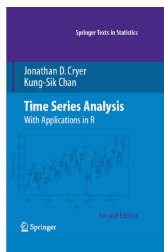
Abderrahim Kessira

Assistant Professor of Statistics

E-mail: abderrahim.kessira@ensia.edu.dz

2 Course Information

2.1 Textbooks



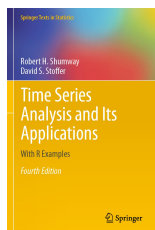
Title: Time Series Analysis with Applications in R (2nd Edition)

Author: J.D. Cryer and K. Chan

Publisher: Springer New York, NY

ISBN: 978-0-387-75958-6

(Primary Textbook)

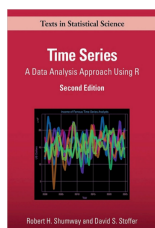


Title: Time Series Analysis and Its Applications

Author: Robert H. Shumway, David S. Stoffer

Publisher: Springer Cham

ISBN: 978-3-031-70586-1



Title: Time Series: A Data Analysis Approach Using R

Author: Robert H. Shumway, David S. Stoffer

Publisher: Chapman and Hall/CRC

ISBN: 978-0-415-00493-0

2.2 Topics Covered

Time Series Analysis:

- Introduction and Examples
- Time Series Models
- Spectral Analysis
- Time-Frequency Representation
- Multivariate Time Series

2.3 Learning Outcomes

By the end of the term successful students should be able to:

- Understand fundamental measures like mean, variance, autocovariance and autocorrelation for time series data
- Understand common time series analysis models and specify and validate appropriate models for real time series data
- Understand the concepts of Fourier time series representation for both univariate and multivariate data as well as wavelet representation.

3 Assessment

1. Continuous Assessment:

- Attendance: 10%
- Midterm: 20%
- Individual Project: 10%

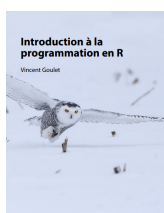
2. Final Exam:

- Group Project: 20%
- Final exam: 40%

4 Computing

The main computing language used for the course is R. All course codes and problem sheets will be illustrated with R. No prior experience with R is assumed.

Please ensure that you install R on your PCs (available at <https://cran.r-project.org/>). You may also install RStudio for a pleasant GUI to work with R (available at <https://posit.co/download/rstudio-desktop/>). We will mainly use the "TSA" and "astsa" packages available for download from the CRAN repositories.



Title: Introduction à la programmation en R

Author: Vincent Goulet

ISBN: 978-2-9811416-6-8

Available at: https://cran.r-project.org/doc/contrib/Goulet_introduction_programmation_R.pdf
